

Ali Sufiyan Khan

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Education

Vellore Institute of Technology (VIT Vellore), B.Tech in Computer Science and Engineering Aug 2023 – May 2027 (Expected)

- **Relevant Coursework:** Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Database Management Systems, Artificial Intelligence

Certifications

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional Aug 2025

- Issued by Oracle University | Valid through August 2027

Projects

Hybrid Multi-Cloud Kubernetes Orchestration Platform github.com/sting-raider/hybrid-cloud-k8s-platform

- Designed a hybrid multi-cloud cluster unifying On-Premise (Edge), AWS, and GCP nodes into a single logical control plane using Kubernetes (K3s) and Tailscale.
- Implemented automated cloud bursting to offload traffic from local infrastructure to public clouds during high-load events, ensuring zero downtime.
- Engineered a global networking layer using Cloudflare Tunnels to expose internal microservices to the public internet securely without port forwarding.
- Conducted distributed load testing with Locust, sustaining 1,000+ concurrent users with 0% failure rate across distributed nodes.

F1 Reinforcement Learning Agent github.com/sting-raider/f1-rl-agent

- Engineered a high-performance autonomous racing agent for F1 22 using Python and Stable Baselines3 (PPO), utilizing an MLP policy on raw telemetry data.
- Developed a low-latency UDP packet parser using Python socket and struct to decode game physics (60Hz) directly.
- Designed a custom Gymnasium environment with a "Virtual Lidar" system to calculate lateral track deviation and lookahead trajectory.
- Implemented continuous action space control via a virtual Xbox 360 controller driver (vgamepad) for precise analog steering.

Context-Aware RAG Agent github.com/sting-raider/AI-ML-PROJECT

- Built and trained a Convolutional Neural Network (CNN) for image classification on the CIFAR-10 dataset, implementing data augmentation.
- Developed an AI assistant using the LangChain framework, integrating Retrieval-Augmented Generation (RAG) to answer questions from a local document knowledge base.

Technologies

Cloud & DevOps: Kubernetes, Docker, AWS (EC2), Google Cloud Platform (GCP), Linux (Bash), Portainer, Netdata

Languages: Python, C, Java, SQL

AI/ML & Frameworks: PyTorch, TensorFlow, LangChain, Scikit-learn, Stable Baselines3, OpenCV

Developer Tools: Git & GitHub, JetBrains IDEs, MySQL